FULL NAME: LEBOGANG SHILWANE
COURSE NAME: DATA SCIENCE
BRIGHTLEARN EXERCISE 2 SQL AGGREGATES AND OPERATORS

1. SELECT DISTINCT department
   FROM students;

department
IT
HR
finance

2. SELECT department,
          AVG(age) AS avg_age
   FROM students;
   GROUP BY department;

department	avg_age
IT	21
HR	22
Finance	23

3. SELECT department,
          COUNT(student_id) AS student_count
FROM students
GROUP BY department
HAVING COUNT(student_id) > 1;

department	studen_count
IT	2
HR	2

4. SELECT student_id,
name,
age,
department
FROM students
WHERE age BETWEEN 21 AND 23;

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	finance
5	Eve	22	HR.

5. SELECT student_id,
name,
age,
department
FROM students
WHERE department IN('IT', 'HR') AND age > 21;

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6. SELECT department,
sum(credits) AS total_credits
FROM courses
GROUP BY department
HAVING sum(credits) > 5;

department	total_credits
IT	11

7.
```
SELECT course_id,
       course_name,
       department,
       credits
FROM courses
WHERE NOT credits = 4;
```

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8.
```
SELECT course_id,
       course_name,
       credits
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

9.
```
SELECT MAX(grade) AS max_grade,
       MIN(grade) AS min_grade,
       AVG(grade) AS avg_grade
FROM enrollments;
```

max_grade	min_grade	avg_grade
90	78	84,6

10. ~~SELECT department,~~
~~sum(salary) AS total_salary,~~
~~sum(bonus) AS total_bonus~~
~~FROM salaries~~
~~GROUP BY department;~~

```
10. SELECT course_id,
           COUNT(enrollment_id) AS enrollment_count
FROM enrollments
GROUP BY course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

```
11. SELECT department,
           sum(salary) AS total_salary,
           sum(bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

department	total_salary	total_bonus
IT	122000	10500
HR	10900	7500
finance	70000	6000

12.
```
SELECT department,
       AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;
```

department	avg_salary
IT	61000
finance	70 000

13.
```
SELECT employee_id,
       name,
       salary,
       bonus,
sum(salary) + sum(bonus) AS total_compensation
FROM salaries
GROUP BY employee_id
HAVING total_compensation > 60 000;
```

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

14.
```
SELECT department,
       sum(budget) AS total_budget,
       AVG(budget) AS avg_budget
FROM projects
GROUP BY department
HAVING avg_budget > 70 000;
```

department	total_budget	avg_budget
IT	270000	135000
Finance	80000	80000

```
~~15. SELECT project_id,~~
~~          project_name,~~
~~          department,~~
~~          budget~~
~~FROM projects~~
~~WHERE budget BETWEEN 50000 AND 120000~~
```

```
15. SELECT project_id,
           project_name,
           department,
           budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000 AND NOT department = 'marketing';
```

Project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll system	Finance	80000
~~5~~	~~HR portal~~	~~HR~~	
5	HR Portal	HR	50000